# Audit Checklist, External Beam Radiation Therapy

Companion table to the article "Audit Tool for External Beam Radiation Therapy Departments" by Ritter, Balter, Lee, Roberts, and Roberson, *Practical Radiation Oncology*, 2012.

## 1.0 Revision history

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| 001 | 15 Nov 2011 | TR, JB, CL, DR, PR | Initial Development. Forward suggestions and questions to Quality Focus Group. |
| 002 | 20 Feb 2012 | TR, JB, CL, DR, PR | Changes per reviewers, added details on respiratory motion, added SRS, added column I. |

## 2.0 Purpose and scope of document

Department Chiefs, administrators, and physicists will use these guidelines to determine if their clinical practices meet the recommended quality and documentation standards.

## 3.0 Definitions

AAPM - American Association of Physicists in Medicine
ACR - American College of Radiology
CAX - Central Axis
CMD - Certified Medical Dosimetrist
CQI - Continuous Quality Improvement
CT - Computed Tomography
DVH - Dose Volume Histogram
I - Importance relative to quality/safety (3 is highest, 1 is lowest)
MU - Monitor Unit
PR - Primary Responsibility
QA - Quality Assurance
SBRT - Stereotactic Body Radiation Therapy
TPS - Treatment Planning System
ADCL - Accredited Dosimetry Calibration Laboratory
ARRT - American Registry of Radiologic Technologists
CBCT- Cone Beam Computed Tomography
CME - Continuing Medical Education
CRCPD - Conference of Radiation Control Program Directors
CTV - Clinical Target Volume
IGRT - Image Guided Radiation Therapy
IMRT - Intensity Modulated Radiation Therapy
OAR - Organ at Risk
PTV - Planning Target Volume
RPC - Radiological Physics Center
TG - Task Group
VMAT - Volumetric Modulated Arc Therapy

## 4.0 Procedure

4.1 Quality procedures and documented records are compared against the following table.

| Item | PR | I | Standard / Criteria | Refs | Frequency |
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| 1 | Admin | 3 | Authorized User / Physician qualifications appropriate and comply with state regulations. Identify written proof of privileges and state licensure. Meet ACR requirements of board certification (or equivalence with completion of residency) and compliance with ACR CME policy (75 cat 1 and 75 cat 2 CME's every 3 years). | 9, 29 | Ongoing |
| 2 | Admin | 3 | Physicist and dosimetrist qualifications appropriate and comply with state regulations. Physicists meet ACR recommendation of board certification + ACR CME policy. Dosimetrists meet ACR recommendation of CMD or certification in progress. | 9, 14, 29 | Ongoing |
| 3 | Admin | 3 | Operator qualifications appropriate and meet ACR recommendation of ARRT (T) and training program meeting CRCPD guidelines. | 9, 29 | Ongoing |

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| 4 | Admin | 1 | Reporting and organizational hierarchy for department documented and appropriate. | - | Ongoing |
| 5 | Admin | 3 | Safety procedures posted at equipment console and appropriate, including emergency contact names and numbers. Staff familiar with content. | 29 | Ongoing |
| 6 | Admin | 2 | Appropriate and detailed method for reporting and maintaining records of medical events/misadministration. Records reviewed for compliance with policy. | 29 | Ongoing |
| 7 | Admin | 2 | Time out checklist before each treatment to verify correct patient, plan, and site. | 36 | Daily |
| 8 | Admin | 3 | All staff empowered and encouraged to stop a procedure if quality or safety is in question. | 36 | Ongoing |
| 9 | Admin | 1 | Documented policies or protocols adequate, spot review of policies on quality management, radiation safety, employee safety, patient safety, pregnant patients, medical events, infection control, imaging and image guidance, treatment planning, SBRT, IMRT, implanted cardiac devices, accelerator QA, CT simulator QA, TPS QA, monitor unit calculations, treatment and prescription records, and competency assessment. | - | Reviewed annually |
| 10 | Admin | 1 | Training and renewed competencies adequate, spot review of competencies for accelerator operation, accelerator daily QA, image guidance, SBRT, IMRT, VMAT, CT simulator operation and QA, treatment planning system operation, emergency treatments, motion management system, radiation safety, patient and employee safety, and emergency treatment procedures. | - | Renewed biennially |
| 11 | Admin | 1 | Policies and procedures:<br>1) are controlled so that only the latest, formally reviewed and approved revision is readily accessible<br>2) are uniquely titled/identified and have a revision date, effective date, and expiration date as needed<br>3) list the primary author and approving authority<br>4) contain a revision history, purpose and scope (or similar categories)<br>5) describe equipment, frequency of required actions/tests, responsible parties, notification paths, and action levels | - | Ongoing |
| 12 | Admin | 1 | Clinical directive, worksheets, and checklists:<br>1) are controlled so that only the latest, formally reviewed and approved revision is readily accessible<br>2) are uniquely titled/identified and have a revision date, effective date<br>3) include spaces for the date and patient information (if applicable)<br>4) include spaces where the individuals completing and approving the document can sign | - | Ongoing |
| 13 | Admin | 1 | Commissioning documents for major new hardware, software, and modalities:<br>1) include a written plan and scope of clinical implementation addressing professional society reports, manufacturer's acceptance testing, manufacturer's recommendations, and relevant literature as appropriate<br>2) document policies and procedures developed during commissioning, including quality assurance methods.<br>3) include a record of relevant training that was performed<br>4) elucidate test equipment, dates, methods, results, and name of person performing each test<br>5) were approved and accepted by the director of clinical physics and/or the medical director of radiation oncology<br>6) were reviewed and approved by the CQI committee prior to clinical implementation<br>7) are assembled, clearly labeled, and kept in one location<br>8) document a review by a designated outside expert for new clinical modalities and procedures (whenever possible)<br>9) summarize follow-up items and demonstrate appropriate follow-up | - | Ongoing |
| 14 | Admin | 1 | Training and competency records:<br>1) demonstrate that training was performed by a competent expert<br>2) include a written lesson plan or summary<br>3) incorporate a method for assessing competency<br>4) include the date accomplished, the name(s) of the trainer(s) and trainee(s), and an expiration date<br>5) are kept current via retraining at appropriate intervals | - | Ongoing |
| 15 | Admin | 2 | CQI committee meetings scheduled and documented, with minutes in place. Medical director has overall responsibility. Committee includes a radiation oncologist, physicist, therapist, and administrative officer (at a minimum). | 1, 9, 35 | Ongoing |

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| 16 | Admin | 1 | Team or individual identified to address recommendations of the Joint Commission on Accreditation of HealthCare Organizations, reports to CQI committee. | - | Ongoing |
| 17 | Admin | 2 | CQI committee reviews new treatment methods and modalities. | 9 | Ongoing |
| 18 | Admin | 2 | CQI committee reviews variances, incidents and accidents (to include any with patient injury), all dose deviations from the prescription greater than 10%, unplanned interruptions during treatment, unusual/severe complications of treatment, and unexpected deaths. | 1, 9 | Ongoing |
| 19 | Admin | 2 | Facility has a standardized method of documenting and acting on variances as part of CQI. | 1 | Ongoing |
| 20 | Admin | 2 | Chart rounds, new patient rounds, morbidity and mortality conference, and monthly chart audit part of CQI program. | 1, 9, 35 | Ongoing |
| 21 | Admin | 2 | Physician peer review program in place with appropriate follow-up. | 9 | Ongoing |
| 22 | Admin | 2 | Patient related outcome data tracked via tumor registry or similar function. | 9 | Ongoing |
| 23 | Admin | 3 | Image guidance and imaging policies in place. Portal images performed for new fields. Setup fields repeated at least every 5 fractions, with margin and delivery specific image guidance policy that includes action levels for shifts. Images labeled with patient name, date, field size, beam, and radiation oncologist approval/date. | 1, 9, 11 | Ongoing |
| 24 | Admin | 2 | Minimum standards for IMRT/VMAT chart documentation met for internally and externally communicated treatment records. Charts should include at a minimum a detailed treatment planning worksheet or directive, DVHs of PTVs/CTVs/OARS, D95, V100, image guidance summary with details of motion management, detailed plan printout to include sagittal and coronal cuts (through PTV(s) and critical structures). | 37 | Each patient |
| 25 | Chief (M.D.) | 2 | Comprehensive history and exam in the chart includes past history, present illness, review of systems, review of imaging studies and lab data, pathology report, and treatment recommendations. | 35 | Ongoing |
| 26 | Chief (M.D.) | 3 | Informed consent appropriate and complete for each patient. Special circumstances, such as treatment of minors, emergency treatments, and IRB approved research study specific consents, are in place. | 38 | Ongoing |
| 27 | Chief (M.D.) | 3 | Rx signed before treatment and includes site, description of portals, modality, energy, dose per fraction, # of fractions per day and week, total # of fractions, total tumor dose, and prescription point or isodose volume. | 1, 9 | Ongoing |
| 28 | Chief (M.D.) | 2 | Physician weekly review of patient and treatment is performed and documented. | 1, 9 | Ongoing |
| 29 | Chief (M.D.) | 2 | Physician assesses tumor response at completion of treatment and documents follow-up plan. Follow-up plan acted on and ongoing visits documented in patient chart. | 1, 9, 35 | Ongoing |
| 30 | Physics | 2 | Radiation safety policy appropriate. Staff can identify and are familiar with radiation safety program documentation. Staff wearing ID badges and dosimeter badges (if likely to receive >10% of dose limit or entering high radiation area). | 29 | Ongoing |
| 31 | Physics | 2 | Record of acceptance testing, commissioning, and periodic QA checks maintained for each accelerator and simulator (CT, MRI or conventional). Acceptance testing must verify manufacturer's specifications and set baseline values of performance. Acceptance and commissioning records maintained at least until machine no longer used or registered with state (consult state or hospital administration for additional guidelines). | 1, 14, 29 | Annual, Monthly, Weekly, Daily |
| 32 | Physics | 2 | Review of accelerator QA program using Tables I – VI in AAPM TG Report 142 on accelerator QA. Ensure IMRT and SRS/SBRT tolerances addressed as indicated. | 18 | As required |
| 33 | Physics | 2 | A QA policy or procedure is in place for all imaging equipment. List equipment and perform spot review of procedures and records. | 14, 18 | As required |
| 34 | Physics | 2 | CT Simulator QA performed per AAPM TG 66 report. Review procedure and test results to ensure compliance with Tables I, II, and III in report. | 22 | As required |
| 35 | Physics | 2 | Imaging dose estimates are available for all clinically used imaging modalities. Techniques or doses are compared to published or manufacturer's baseline values and reviewed for appropriateness. Data used to reduce dose and eliminate un-needed exposures. | 19 | Annually |
| 36 | Physics | 2 | Records of maintenance for major equipment include name of person performing service, date, and signature of physicist authorizing return to service. | 29 | Ongoing |
| 37 | Physics | 2 | Shielding design approved and drawings/design available. Must include assessment for IMRT. | 12, 29, | Acceptance, |

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| (cont) | | | | 39 | after changes. |
| 38 | Physics | 2 | Shielding survey records reviewed. Primary, scatter, and leakage radiation measured at acceptance and after changes in barriers or workload. Appropriate workloads used to demonstrate compliance with NCRP 151 and state guidelines. | 29, 39 | Acceptance, after changes. |
| 39 | Physics | 3 | Calibrated local standard ion chamber dosimetry system available with current ADCL certificate. Ion chamber calibrated biennially. Calibrated survey instrument available, 10 uSv/hr to 10 mSv/hr range. Survey instrument calibrated annually. | 1, 14, 29 | Ongoing |
| 40 | Physics | 2 | Program to ensure accuracy of measurement equipment to include ion chambers, barometers, thermometers, and IMRT QA device(s). | 1, 14, 26 | Annual, monthly |
| 41 | Physics | 3 | Wedges, cones, and filters have appropriate labels and interlocks. Field sizes exceeding wedge, cone, or filter dimensions are interlocked. | 1, 29 | Ongoing |
| 42 | Physics | 2 | Output calibration of linear accelerator in compliance with TG-51 report. Calculation verified for recent calibration. | 20 | Annual |
| 43 | Physics | 3 | Independent check of output calibration of linear accelerator performed annually. | 14 | Annual |
| 44 | Physics | 3 | Control panel check – lock out used and key secured. Warning light, beam off, door interlock, camera system, and emergency off all functional and checked regularly. | 1, 9, 18, 29 | Ongoing, Daily and Monthly |
| 45 | Physics | 2 | IMRT/VMAT commissioning report includes delivery system and TPS verifications. IMRT commissioning based on AAPM Reports 82 and 119. Staff training/competencies and written procedures verified. QA equipment in line with AAPM TG Report 120. | 23,25, 26, 29 | Commissioning |
| 46 | Physics | 2 | Physicist has a documented, detailed procedure for checking a treatment plan and dose calculation. Compare to pages 609 – 610 of TG-40 report. | 1, 14 | Ongoing |
| 47 | Physics | 3 | Patient specific IMRT/VMAT QA includes, in addition to standard treatment plan check, verification of correct transfer of plan and MLCs to record and verify system and patient specific end-to-end testing to validate dose distribution and delivered dose. QA must be performed and approved before 1st fraction. IMRT/VMAT QA approved by physician. Confidence limits and action levels appropriate as described in AAPM Report 119. | 12, 25 | Each patient |
| 48 | Physics | 3 | Check of plan and dose calculations, to include independent second check of the MUs, accomplished prior to 3rd fraction / 10% of dose delivered OR prior to 1st fraction if 5 or fewer total fractions. Both physicist and physician approval verified. | 1, 9, 14 | Ongoing |
| 49 | Physics | 2 | Procedure in place for calculating monitor units based on measured parameters. Dosimetry parameters used for calculation readily available. | 1, 14 | As required. |
| 50 | Physics | 3 | Physicist performs weekly chart review of treatment prescription, MU calculations, daily treatment records, any new or modified treatment fields, and cumulative doses. Physicist participates in weekly chart rounds. | 1, 14 | Weekly |
| 51 | Physics | 2 | Physicist performs final chart review for fulfillment of prescription dose, initial/date indicate review within 1 week of treatment completion. | 1, 14 | As required. |
| 52 | Physics | 2 | Annual program review as part of quality management program, including review of physics and radiation safety programs, reported to CQI committee. | 1, 14, 16 | Annual |
| 53 | Physics | 2 | Outside review (TG-103 or similar peer review, accreditation, RPC site visit, or other formal audit by a qualified physicist) accomplished within last 14 months. Evidence of follow-up should be documented. | 16 | Annual |
| 54 | Physics | 3 | Rigorous treatment planning system acceptance and commissioning performed and documented. All used features tested to include CAX and off-axis points, open fields, blocked fields, wedged fields, small and large field sizes. Methods of normalization and absolute doses verified. Dose distributions and DVHs validated. Coordinate system tests performed and validated via end-to-end test. | 14, 21 | Commissioning |
| 55 | Physics | 2 | Treatment planning system (TPS) QA performed and documented. Weekly check of machine files and directories. Monthly tests include configuration and error review, check of major input (CT geometry and electron density), and output. Annual check of all input (MR, PET), repeat of reference test sets for each algorithm, and check of critical software tools. Upgrades and changes tested and clinically released. | 1,14, 21, 40 | Weekly, Monthly and Annual |
| 56 | Physics | 2 | Log of TPS failures, errors, changes, and upgrades maintained. | 10 | Ongoing |
| 57 | Physics | 2 | Treatment planning system training performed and documented for all users. Training for new releases is documented. | 14 | As required |
| 58 | Physics | 3 | Record and verify functionality confirmed (to include transfer from TPS, transfer to treatment unit, saving from treatment unit). | 14 | As required. |
| 59 | Physics | 2 | Respiratory motion management system (i.e. motion encompassing, gating, breath hold, abdominal compression, or tumor tracking) implemented. Appropriate clinical process, treatment planning, and QA procedures documented and followed as described in the report of TG-76. If respiratory gating used, then QA performed per AAPM TG-142. Radiograph of tumor/surrogate, or other validation method, | 11, 18, 27 | At initiation of program and ongoing |

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| 59 (cont) | | | used to check internal constancy at patient setup. Staff training and competencies accomplished and documented. | | |
| 60 | Physics | 3 | SBRT program review of commissioning report, documented staff training, and TPS validations tests to include tissue inhomogeneity corrections and small fields. Written procedures address staff responsibilities, simulation, treatment planning, OAR dose limits, treatment imaging and delivery, verification checks, and QA. Entire treatment chain validated in end-to-end test. | 13,17, 41 | At initiation of program and ongoing |
| 61 | Physics | 3 | SBRT treatments follow established guidelines. QC checklist(s) developed by physicist guide entire process. Medical physicist present for first fraction. Radiation oncologist approves image guidance prior to treatment and directs entire treatment. All treatment parameters verified immediately prior to delivery. Redundant check of patient geometric alignment used per AAPM TG 101 (SBRT). | 13,17, 41 | Each fraction |
| 62 | Physics | 3 | SRS program review of commissioning report, documented staff training, TPS validations tests to include tissue inhomogeneity corrections and small fields (with secondary verification), localization test results, mechanical precision of frame and docking system, target and beam alignment test, and safety interlocks for collimators/couch/gantry/cones. Written procedures address staff responsibilities, simulation, treatment planning, OAR dose limits, treatment imaging and delivery, verification checks, and QA. | 15, 28, 41 | At initiation of program and ongoing |
| 63 | Physics | 3 | SRS treatments follow established guidelines. QC checklist(s) developed by physicist guide entire process. Checks prior to treatment include verification of 2mm or smaller grid size for calculations, plan dose gradients > -60%/3mm, correct registration and geometric accuracy of secondary datasets, correct localization frame coordinates, and correct isocenter coordinates. Appropriate neurosurgery involvement confirmed. Patient specific end-to-end test performed each case. Medical physicist present for each fraction. Radiation oncologist directs entire treatment, verifies site, and approves setup and image guidance. Redundant check of patient alignment used. All treatment parameters verified immediately prior to delivery. | 15, 28, 41 | Each fraction |
| 64 | Physics | 1 | Site has a centralized QA tracking method (software, spreadsheet, calendar, other) to ensure all QA performed at appropriate intervals. | - | Ongoing |
| 65 | Physics | 1 | System in place to review and act on vendor notices. | - | Ongoing |

## 5.0 <u>Responsibilities</u>

6.1 Sites should assess their department for compliance with these recommendations. Non-compliances should be compared against the original standard and reviewed with the department quality committee for consideration.

## 6.0 <u>References</u>

The following references are numbered sequentially as in the published manuscript.


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